MIL-C-60898/7 (MU)
31 July 1969

# MILITARY SPECIFICATION

## CIRCUIT CARD ASSEMBLY, RELAY SIGNAL INVERTERS

## 1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-60898 comprise the complete requirements for the Relay Signal Inverters Circuit Card Assembly.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-C-60898 | Circuit Card Assemblies, General Specification for |

DRAWINGS

U. S. Army, Frankford Arsenal

| | |
|---|---|
| F10554189 | Electronic Component Assembly, Relay Time |

PACKAGING DATA SHEET

| | |
|---|---|
| 10554189 | Packaging of Circuit Card Assembly, Relay Signal Invester |

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 4931

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## 3. REQUIREMENTS

3.1 Fabrication. The assembly shall be manufactured in accordance with Drawing F10554189 and drawings pertaining thereto.

3.2 General specifications. The assembly shall meet the requirements of MIL-C-60898.

3.3 Environmental service conditions. See MIL-C-60898.

3.3.1 Performance at temperature extremes. For the conditions of 3.3.2 of MIL-C-60898, the voltage present at the output pins specified in Table I shall be greater than +0.05 vdc but less than +1.2 vdc for the high input of 3.4.1. For the low input of 3.4.2, the voltage at the output pins shall be 20.5 ± 1.5 vdc.

3.4 Electrical performance. The requirements of 3.4.1 and 3.4.2 shall be accomplished with the test set-up shown in Figure 1.

3.4.1 High input. With +22 ± 0.1 vdc applied between the input pins specified in Table 1 and pin 1 (-), a voltage greater than +0.05 vdc but less than +1.2 vdc shall be present between the output pins specified in Table 1 and pin 1 (-).

3.4.2 Low input. With +0.50 ± 0.01 vdc applied between the inputs specified in Table I and pin 1 (-), a voltage of +21.5 ± 0.5 vdc shall be present between the output pins specified in Table I and pin 1 (-).

Table I

Voltage Terminations

| INPUT | OUTPUT |
|---|---|
| 3 | 2 |
| 7 | 5 |
| 9 | 11 |
| 10 | 8 |
| 12 | 15 |
| A* | 16 |
| B* | 18 |
| C* | 20 |
| D* | 22 |

* See Figure 1



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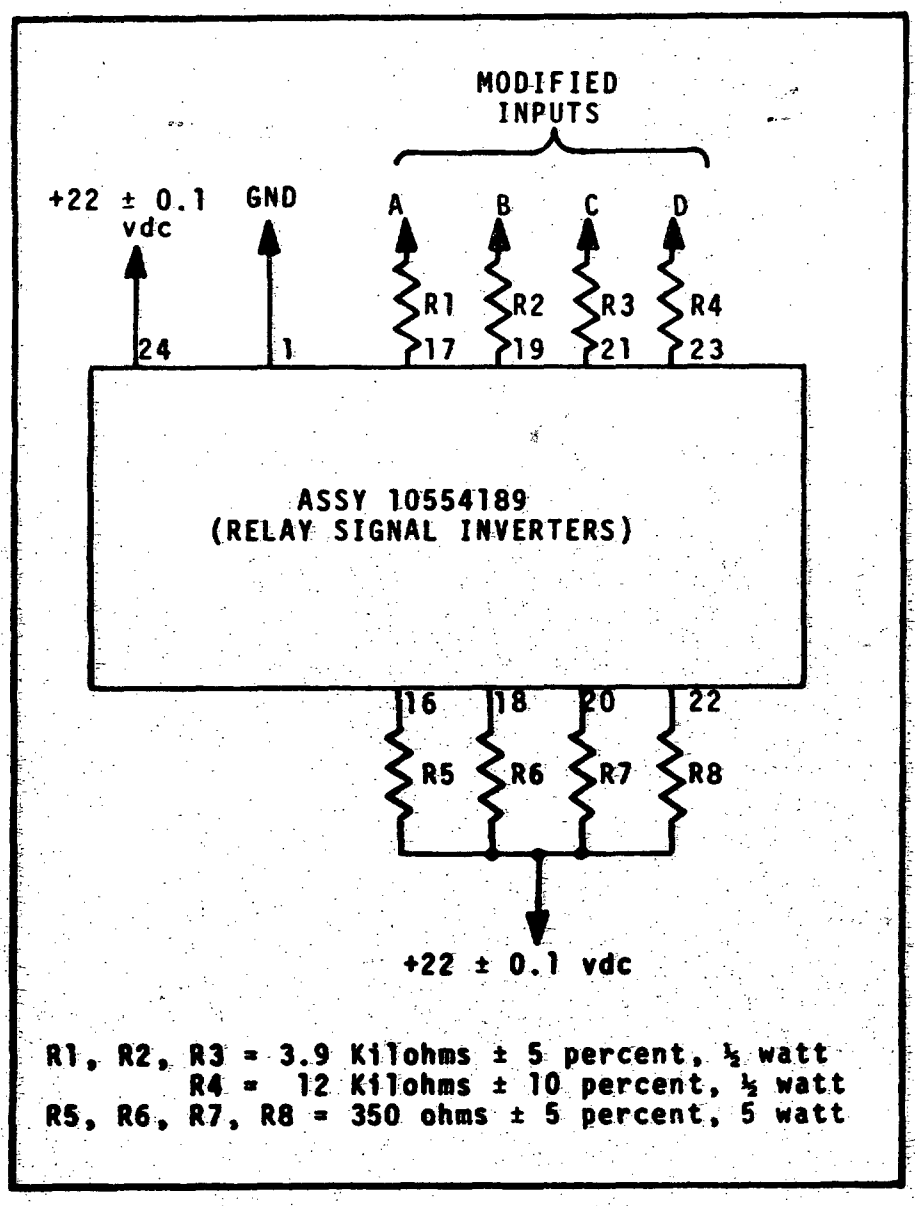



Figure 1



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3.4.3 Capacitance. With the signal shown in Figure 2 applied through a 100 ohm ± 10 percent resistor, between pin 24 (+) and pin 1, the voltage out between pin 13 (+) and pin 1 shall be as shown in Figure 3.

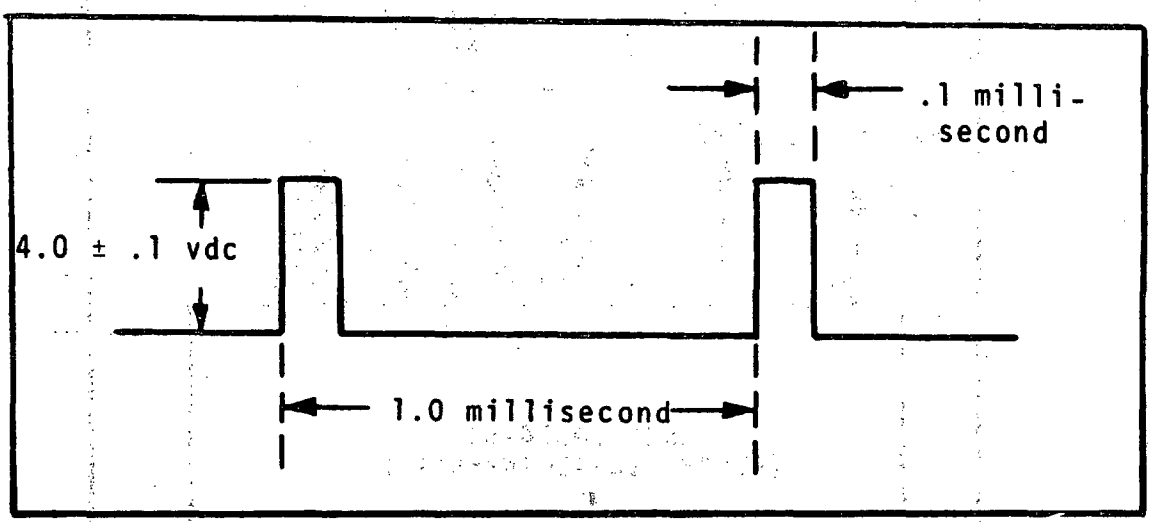



Figure 2

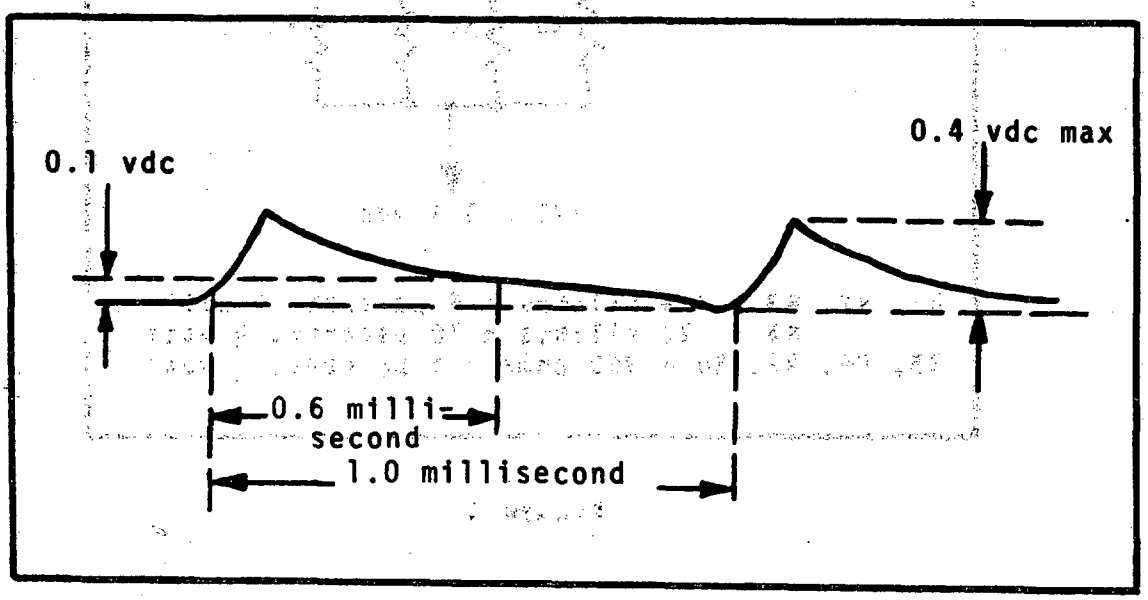



Figure 3



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 General. The quality assurance provisions of MIL-C-60898 shall apply to the assembly covered by this detail specification.

4.2 First article inspection. As specified in MIL-C-60898.

4.3 Control test sample. See MIL-C-60898.

4.4 Functional tests. The following characteristics shall be tested 100 percent in accordance with the applicable test method.

| Characteristic | Requirement | Procedure |
|---|---|---|
| 100 High input | 3.4.1 | 4.5.2.1 |
| 101 Low input | 3.4.2 | 4.5.2.2 |
| 102 Capacitance | 3.4.3 | 4.5.2.3 |

4.4.1 Inspection equipment. As specified in MIL-C-60898 and 4.4.1.1.

4.4.1.1 Test equipment. In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5.

| Item | Description |
|---|---|
| 1. POWER SUPPLY | Nominal value of +22.0 ± 0.1 vdc, no less than 500 ma; ± 1 percent regulation; 0.1 percent maximum ripple. |
| 2. PULSE GENERATOR | Frequency - 1 kilohertz minimum. Duty factor - 10 percent minimum. Amplitude - a minimum of 4 volts into a 200 ohm load. Pulse rise and fall times - less than 0.2 microseconds. |
| 3. POWER SUPPLY | Nominal value of 0.50 ± 0.01 vdc; no less than 50 ma; ± 1 percent regulation; 0.1 percent maximum ripple. |



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| | |
|---|---|
| 4. DIGITAL VOLTMETER | Range of no less than 0.0010 to 999.9 vdc; four significant figures; accuracy of no less than ± 0.2 percent; input impedence no less than 10 megohms on all ranges. |
| 5. DECADE RESISTOR | Range of 100 ohms to 100 kilohms ± 0.1 percent; 1 watt. |
| 6. VARIABLE RESISTOR | Range of 0.0 to 500 ohms ± 0.1 percent; 5.0 watt. |
| 7. OSCILLOSCOPE | Voltage and time accuracy of ± 10 percent; rise time no greater than 100 nanoseconds; sweep period of 1 microsecond to 20 milliseconds or greater; input impedance of no less than 1 megohm. |

4.5 Test Methods and Procedures

4.5.1 Environmental. As specified in MIL-C-60898 except the requirements of 3.3.1 herein shall be verified during temperature extremes.

4.5.2 Electrical tests

4.5.2.1 High input

a. Using POWER SUPPLY, item 1 of 4.4.1.1, apply + 22 vdc between pin 24 (+) and pin 1 (-). Apply the same +22 vdc individually between the input pins specified in Table II (+) and pin 1 (-). Connect the DIGITAL VOLTMETER, item 4 of 4.4.1.1, individually between the corresponding output pins specified in Table II (+) and pin 1 (-). The output voltage indicated on the DIGITAL VOLTMETER shall be between +0.05 and +1.2 vdc.



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TABLE II

HIGH INPUT TERMINATIONS

| Input | Output |
|---|---|
| Pin 3 | Pin 2 |
| 7 | 5 |
| 9 | 11 |
| 10 | 8 |
| 12 | 15 |

b. Using POWER SUPPLY; item 1 of 4.4.1.1, apply +22 vdc between one terminal of the DECADE RESISTOR, item 5 of 4.4.1.1 and one terminal of the VARIABLE RESISTOR, item 6 of 4.4.1.1. Adjust the DECADE RESISTOR to 3.9 kilohms and the VARIABLE RESISTOR to 350 ohms. Connect the free terminals of the resistors individually to the pins listed in Table III. Connect the DIGITAL VOLTMETER, item 4 of 4.4.1.1, individually between the pins showing the Table III (+) and pin 1 (-). The output voltage indicated on the DIGITAL VOLTMETER shall be between +0.05 vdc and +1.2 vdc.

TABLE III

HIGH INPUT TERMINATIONS

| Decade Resistor | Variable Resistor | Digital Voltmeter |
|---|---|---|
| Pin 17 | Pin 16 | Pin 16 |
| 19 | 18 | 18 |
| 21 | 20 | 20 |

c. Using POWER SUPPLY, item 1 of 4.4.1.1, apply +22 vdc between one terminal of the DECADE RESISTOR, item 5 of 4.4.1.1 and one terminal of the VARIABLE RESISTOR, item 6 of 4.4.1.1 Adjust the DECADE RESISTOR TO 12 kilohms and the VARIABLE RESISTOR to 350 ohms. Connect the DIGITAL VOLTMETER between pin 22 (+) and pin (-). Connect the free terminal of the DECADE RESISTOR to pin 23. Connect the free terminal of the VARIABLE RESISTOR to pin 22. The output voltage indicated on the DIGITAL VOLTMETER shall be between +0.05 vdc and +1.2 vdc.

4.5.2.2 Low input

a. Using POWER SUPPLY, item 1 of 4.4.1.1, apply +22 vdc between pin 24 (+) and pin 1 (-). Using POWER SUPPLY, item



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3 of 4.4.1.1, apply +0.5 vdc individually between the input pins shown in Table II (+) and pin 1 (-). Connect the DIGITAL VOLTMETER, item 4, of 4.4.1.1, individually between the output pins of Table II (+) and pin 1 (-). The output voltage indicated on the DIGITAL VOLTMETER shall be +21.5 ± 0.5 vdc.

b. Using POWER SUPPLY, item 1 of 4.4.1.1, apply +22 vdc to one terminal of the DECADE RESISTOR, item 5 of 4.4.1.1. Using POWER SUPPLY, item 3 of 4.4.1.1 apply +0.5 vdc to one terminal of VARIABLE RESISTOR, item 6 of 4.4.1.1. Adjust the DECADE RESISTOR to 3.9 kilohms and the VARIABLE RESISTOR to 350 ohms. Connect the free terminals of the RESISTORS to the terminals shown in Table III. Connect the DIGITAL VOLTMETER, item 4 of 4.4.1.1 between the pins shown in Table III (+) and pin 1 (-). The output voltage, indicated on the DIGITAL VOLTMETER, shall be +21.5 ± 0.5 vdc.

c. Using POWER SUPPLY, item 1 of 4.4.1.1, apply +22 vdc to one terminal of the DECADE RESISTOR, item 5 of 4.4.1.1. Using POWER SUPPLY, item 3 of 4.4.1.1, apply +0.5 vdc to one terminal of VARIABLE RESISTOR, item 6 of 4.4.1.1. Adjust the VARIABLE RESISTOR to 350 ohms and the DECADE RESISTOR 12 kilohms. Connect the DIGITAL VOLTMETER, item 4 of 4.4.1.1, between pin 22 (+) and pin 1 (-). Connect the free terminal of the DECADE RESISTOR to pin 23 and the free terminal of the VARIABLE RESISTOR to pin 22. The output voltage, indicated on the DIGITAL VOLTMETER, shall be +21.5 ± 0.5 vdc.

4.5.2.3 Capacitor C1. Connect the PULSE GENERATOR, item 2 of 4.4.1.1, through the DECADE RESISTOR, item 5 of 4.4.1.1 to pin 24 (+) and pin 1 (-) as shown in Figure 4. Connect the channel A vertical input of the OSCILLOSCOPE, item 7 of 4.4.1.1, to the output of the PULSE GENERATOR and pin 1(-). Connect the channel B vertical input of the OSCILLOSCOPE to pin 24 (+) and pin 1 (-). Adjust the DECADE RESISTOR to 100 ± 10 ohms. Adjust the PULSE GENERATOR to obtain the waveform of Figure 2 on channel A of the OSCILLOSCOPE. Verify the waveform of Figure 3 on channel B of the OSCILLOSCOPE.



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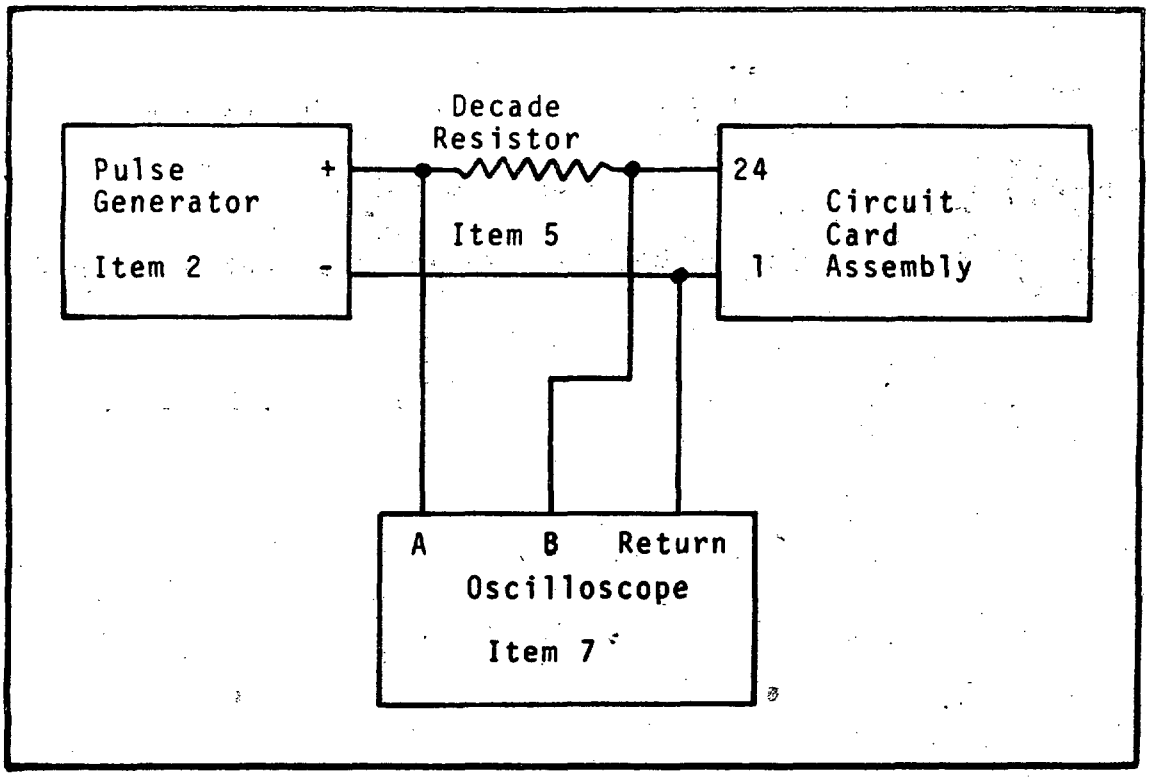



Figure 4



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## 5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with Packaging Data Sheet 10554189.

## 6. NOTES

6.1 The notes specified in MIL-C-60898 are applicable to this specification.

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